# JOTHI PRAKASH N

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## PROFESSIONAL SUMMARY

Computer Science Engineer skilled in Python (Flask), Java, React, and databases, with hands-on experience building full-stack applications, AI-powered systems, computer vision solutions, and embedded projects. Interested in LLMs, local AI, RAG, and intelligent user experiences. Collaborative and self-driven with a strong focus on practical problem solving.

## TECHNICAL SKILLS

Languages: JavaScript, Python, Java
Frameworks: Flask, React
Databases: MySQL, PostgreSQL, SQLite
AI / ML: LLMs, Local LLM (Ollama), RAG, Embeddings, Vector Search, Computer Vision, Face Recognition, Gesture Recognition
Tools & Platforms: Git, GitHub, pgAdmin, Postman, Maven, OpenCV, MediaPipe, PyAutoGUI, Raspberry Pi, Arduino

## PROJECTS

### Fusion Cart — AI-Powered E-Commerce Platform

Python, React, Flask, Ollama, LLM, RAG

- Built a full-stack e-commerce platform where a conversational AI assistant supports product discovery, cart management, and order operations.
- Integrated a local LLM using Ollama to enable natural-language shopping assistance with privacy-focused local inference.
- Designed RAG-based product search using embeddings and vector search to provide context-aware product information.
- Used LLM tool/function calling concepts to connect conversational requests with product search, cart, and order-management actions.
- Developed the React frontend and Flask backend with a client-server architecture for end-to-end application ownership.

### Hand Gesture Recognition System

Python, OpenCV, MediaPipe, PyAutoGUI

- Built a real-time computer vision system that detects hand gestures through a webcam and maps gestures to system volume controls.
- Used MediaPipe to track 21 hand landmarks per frame and implemented gesture classification logic for responsive controls.
- Tuned detection thresholds to reduce false triggers and improve responsiveness under varied lighting conditions.

### AI-Based Face Recognition Smart Door Lock System

Python, OpenCV, Flask, SQLite, Arduino, Raspberry Pi

- Developed a smart door lock combining Raspberry Pi and Arduino for automated access control.
- Implemented real-time face recognition using OpenCV and facial feature matching to identify authorized users.
- Programmed Arduino to control a servo motor, buzzer, and LED indicators based on recognition results.
- Stored user details and access logs in SQLite and built a Flask dashboard to manage users and view access history.
- Integrated embedded hardware with the computer-vision recognition pipeline for an end-to-end intelligent access system.

## EDUCATION

### B.E. in Computer Science and Engineering — Graduated 2026

Study World College of Engineering | CGPA: 7.9

## CERTIFICATIONS

### Java Programming — Qspiders, 2026

- Completed structured training in Core and Advanced Java, covering OOP concepts, collections, exception handling, and multithreading.
- Gained hands-on practice building Java-based applications and solving real-world programming problems.

## CORE COMPETENCIES

- Problem Solving & Debugging — breaking down complex issues into manageable, testable steps.
- Full-Stack Development — comfortable working across frontend, backend, database, and API layers.
- AI Application Development — interested in integrating LLMs, RAG, computer vision, and intelligent automation into applications.
- Rapid Prototyping — quick to build and iterate on working proofs-of-concept across software and hardware projects.
- Team Collaboration — effective communicator who works well in cross-functional, fast-paced environments.
- Self-Directed Learning — proactively picks up new tools and frameworks to complete projects independently.